

Problem

Find the inverse Laplace transform of:

$$F(s) = \frac{s^2 + 2}{s^3 + 2s^2 + 2s}$$

Step-by-step solution

Step 1 of 4

The Laplace transform is,

$$\begin{aligned} G(s) &= \frac{s^2 + 2}{s^3 + 2s^2 + 2s} \\ &= \frac{s^2 + 2}{s(s^2 + 2s + 2)} \end{aligned}$$

Apply partial transforms to simplify the above equation.

$$\frac{s^2 + 2}{s(s^2 + 2s + 2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 2} \dots\dots (1)$$

Here A , B , and C are the constants

$$s^2 + 2 = A(s^2 + 2s + 2) + (Bs + C)(s)$$

$$s^2 + 2 = A(s^2 + 2s + 2) + Bs^2 + Cs \dots\dots (2)$$

Step 2 of 4

Equate the constant terms from equation (2).

$$2 = 2A$$

$$A = 1$$

Equate the s coefficients from equation (2).

$$0 = 2A + C$$

$$C = -2A$$

$$= -2$$

Equate the s^2 coefficients from equation (2).

$$1 = A + B$$

$$1 = 1 + B$$

$$B = 0$$

Step 3 of 4

Recall equation (1).

$$G(s) = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 2}$$

Substitute 1 for A , 0 for B and -2 for C in the above equation.

$$\begin{aligned} G(s) &= \frac{1}{s} + \frac{(0)s + (-2)}{s^2 + 2s + 2} \\ &= \frac{1}{s} - \frac{2}{s^2 + 2s + 2} \\ &= \frac{1}{s} - \frac{2}{(s+1)^2 + 1^2} \end{aligned}$$

Step 4 of 4

It is known that,

$$L^{-1}\left\{\frac{1}{s}\right\} = u(t)$$

$$L^{-1}\left\{\frac{b}{(s+a)^2 + b^2}\right\} = e^{-at} \sin bt$$

The Laplace function is,

$$G(s) = \frac{1}{s} - \frac{2}{(s+1)^2 + 1^2}$$

Apply inverse Laplace transform.

$$\begin{aligned} g(t) &= u(t) - 2e^{-t} \sin(t)u(t) \\ &= [1 - 2e^{-t} \sin(t)]u(t) \end{aligned}$$

Therefore, the inverse Laplace transform of $F(s)$ is $\boxed{[1 - 2e^{-t} \sin(t)]u(t)}$.